

E9: 309 Advanced Deep Learning

28-10-2020

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<http://leap.ee.iisc.ac.in/sriram/teaching/ADL2020/>

Housekeeping

✳ Attendance

✓ We will use the recorded sessions for attendance

★ If you are unable to attend live sessions (due to network or other issues, please indicate by email before or after class to the instructor and copy the FAs).

✳ Mid-term exam

➡ 1st week of Dec. (Modules I and II).



Housekeeping

✦ 1st mini-project

✓ Deadlines

- ★ Abstract submission deadline (Nov 2nd, Monday)
 - ★ Using the google form given in the webpage
- ★ Solo projects or 2-member projects
 - ★ Indicate roles of each member in 2-member project
 - ★ 200 word abstract of the work. If modifications are needed, we will review and let you know in 2-3 days.



Housekeeping

✳ 1st mini-project

✓ Deadlines

- ★ Report and presentation slides (Nov 19th, 10 AM).
 - ★ 1-page pdf with second page only for references and tools used (Template will be provided).
- ★ Report - Indicate prior work, technical details and your contribution. Strictly adhere to the guidelines given in the template.
- ★ Slides (max 4 slides) - 4 min presentation for solo project and 6 min. for two member teams. 3 mins for your presentation and 1 min for Q&A.
- ★ Two slots are available on 2 days (pick the suitable based on your other class schedules).



Recap of previous class



State of affairs

✳️ Encoder-decoder models.

- ✓ Issues with single encoder embedding for all outputs

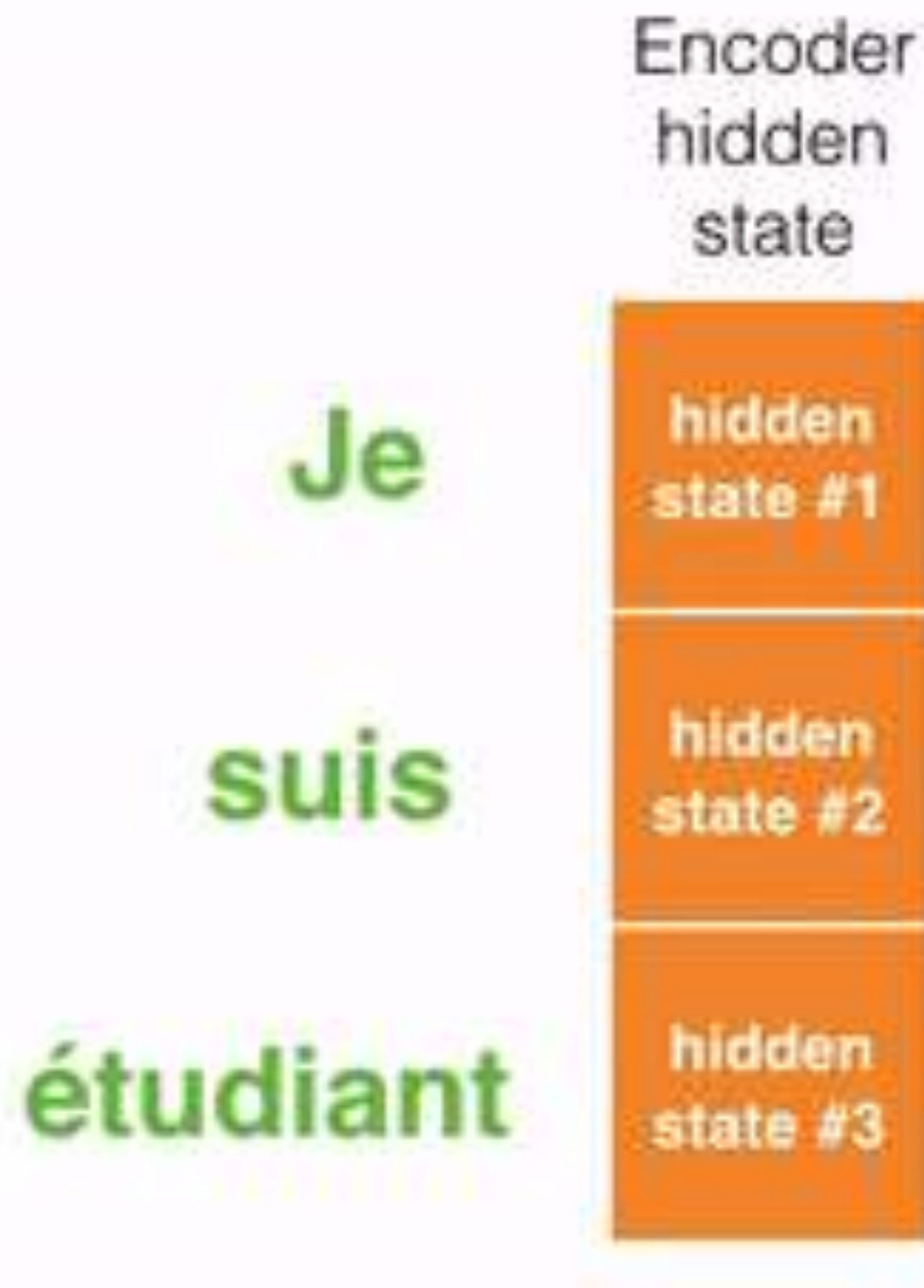
➡ Introduction to attention

- ✓ Attention network and attention weights

➡ Visualizing attention weights.

➡ Self-attention and multi-head attention.,

Visualizing attention



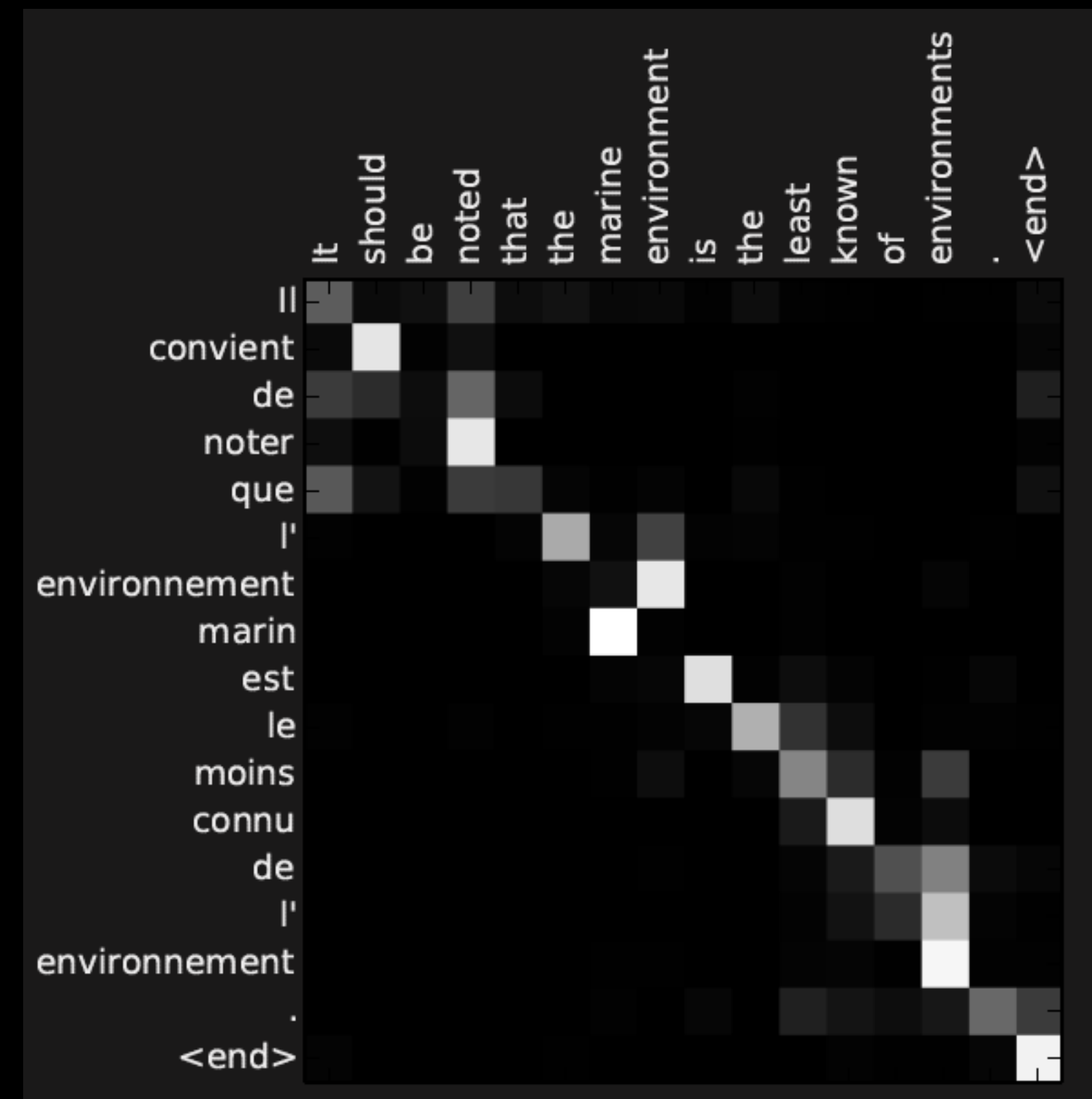
Analysis of attention networks

✳ Attention weights $\alpha(s, t)$

- ✓ Probability of linking (attending) to input at t for generating output at s
- ✓ Useful in analyzing the internal structure of the encoder-decoder model

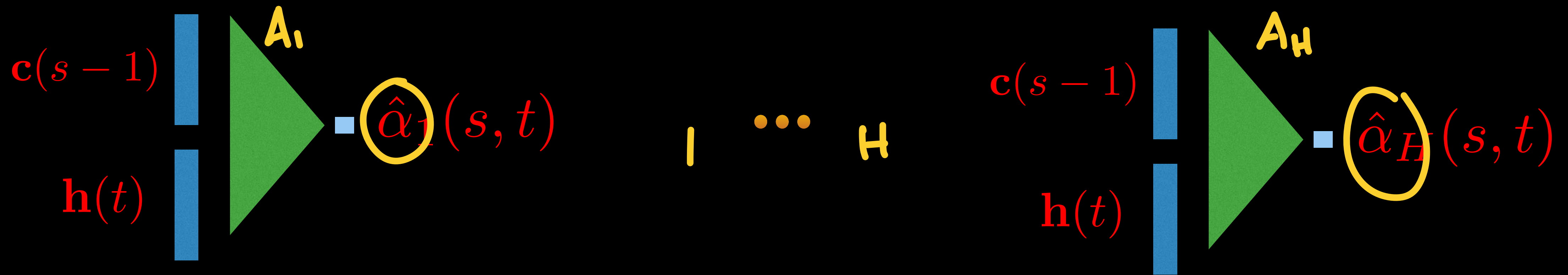
Visualizing the attention weights

Reading Assignment - “Neural Machine Translation by Jointly Learning to Align and Translate”
<https://arxiv.org/pdf/1409.0473.pdf>



Multi-head attention

✳ Having more than one attention heads



$$\hat{\alpha}_1(s, t) = \mathbf{A}_1[\mathbf{c}(s-1); \mathbf{h}(t)]$$

$$\hat{\alpha}_H(s, t) = \mathbf{A}_H[\mathbf{c}(s-1); \mathbf{h}(t)]$$

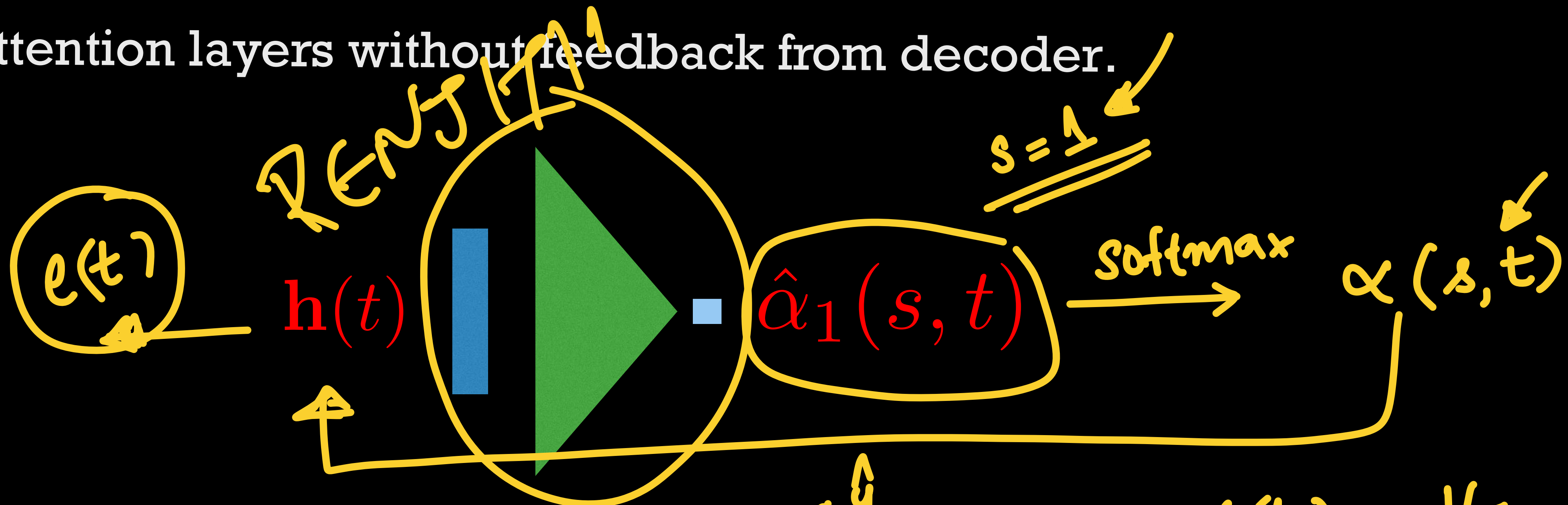
$$\mathbf{e}_1(s) = \sum_t \alpha_1(s, t) \mathbf{h}(t)$$

$$\mathbf{e}_H(s) = \sum_t \alpha_H(s, t) \mathbf{h}(t)$$

$$\mathbf{e}(s) = [\mathbf{e}_1^T(s); \dots; \mathbf{e}_H^T(s)]^T$$

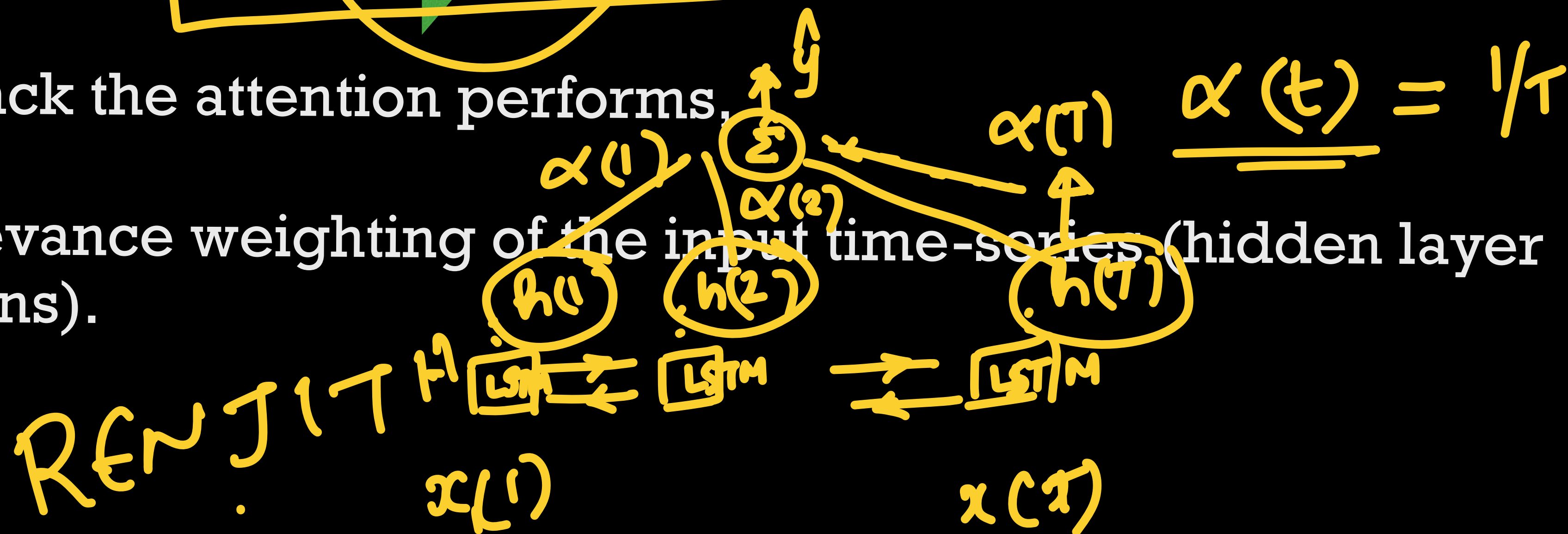
Self-attention

- * Using attention layers without feedback from decoder.



- * Without feedback the attention performs,

- temporal relevance weighting of the input time-series (hidden layer representations).



Issues in RNNs/LSTMs

- ✱ Issues of long-term dependency

- ➡ LSTMs have partial solutions

- ✱ Back propagation through time

- ➡ Does not allow parallelism in forward pass or backward pass.

- ➡ Significant increase in training time as well as in forward propagation.

- ✱ **Question** - can we use attention mechanism itself to build temporal dependencies without recurrence.



Transformers

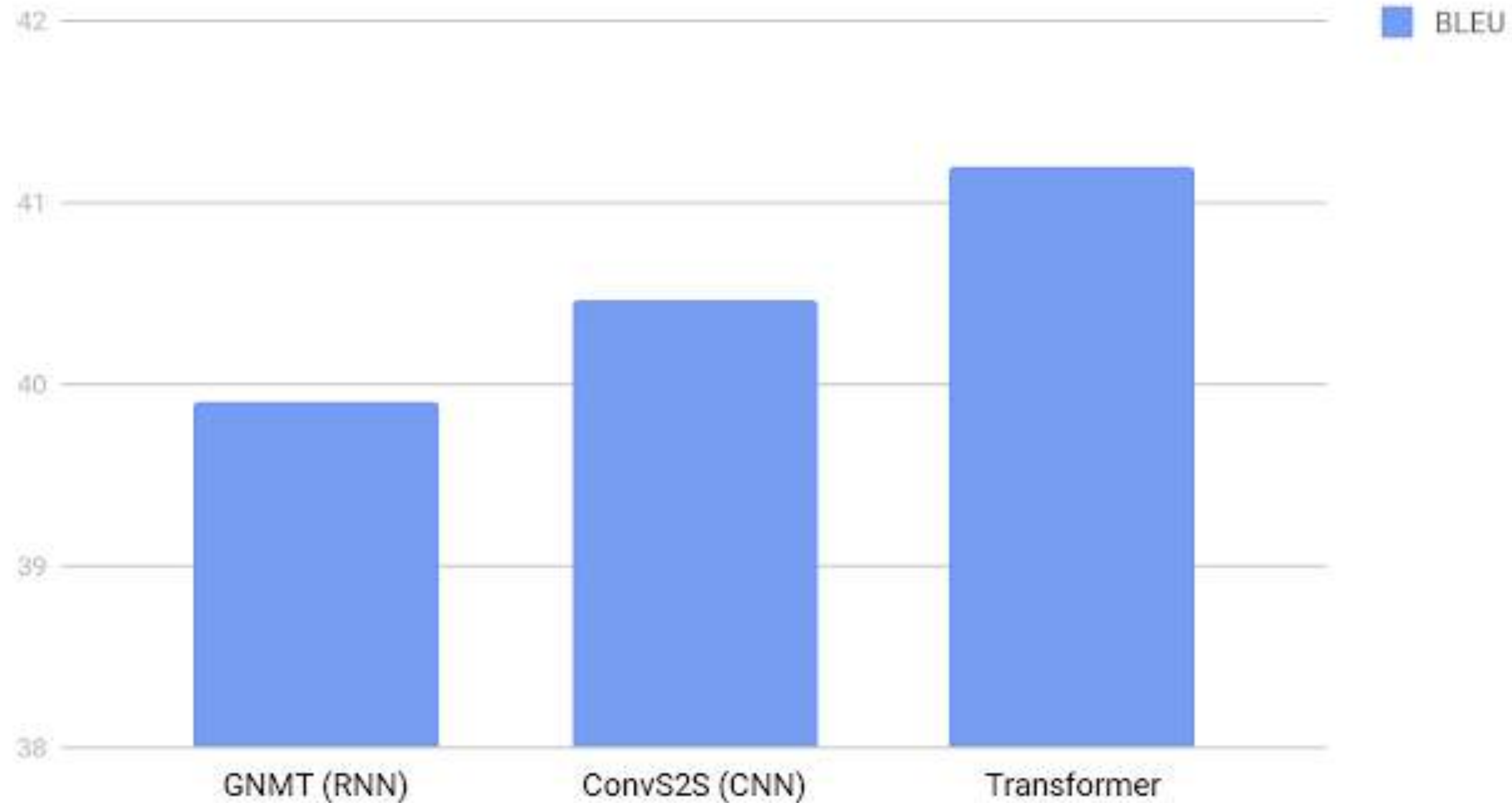
- * Encoder Decoder architecture based models.
- * Uses only feed forward architectures with self-attention.
- ➔ Multi-head self attention.
- * All the encoder layers and the decoder layers have the same set of operations.

Reading Assignment - "Attention is All You Need"
<https://arxiv.org/pdf/1706.03762.pdf>



Transformers - the state of art in NMT

English French Translation Quality



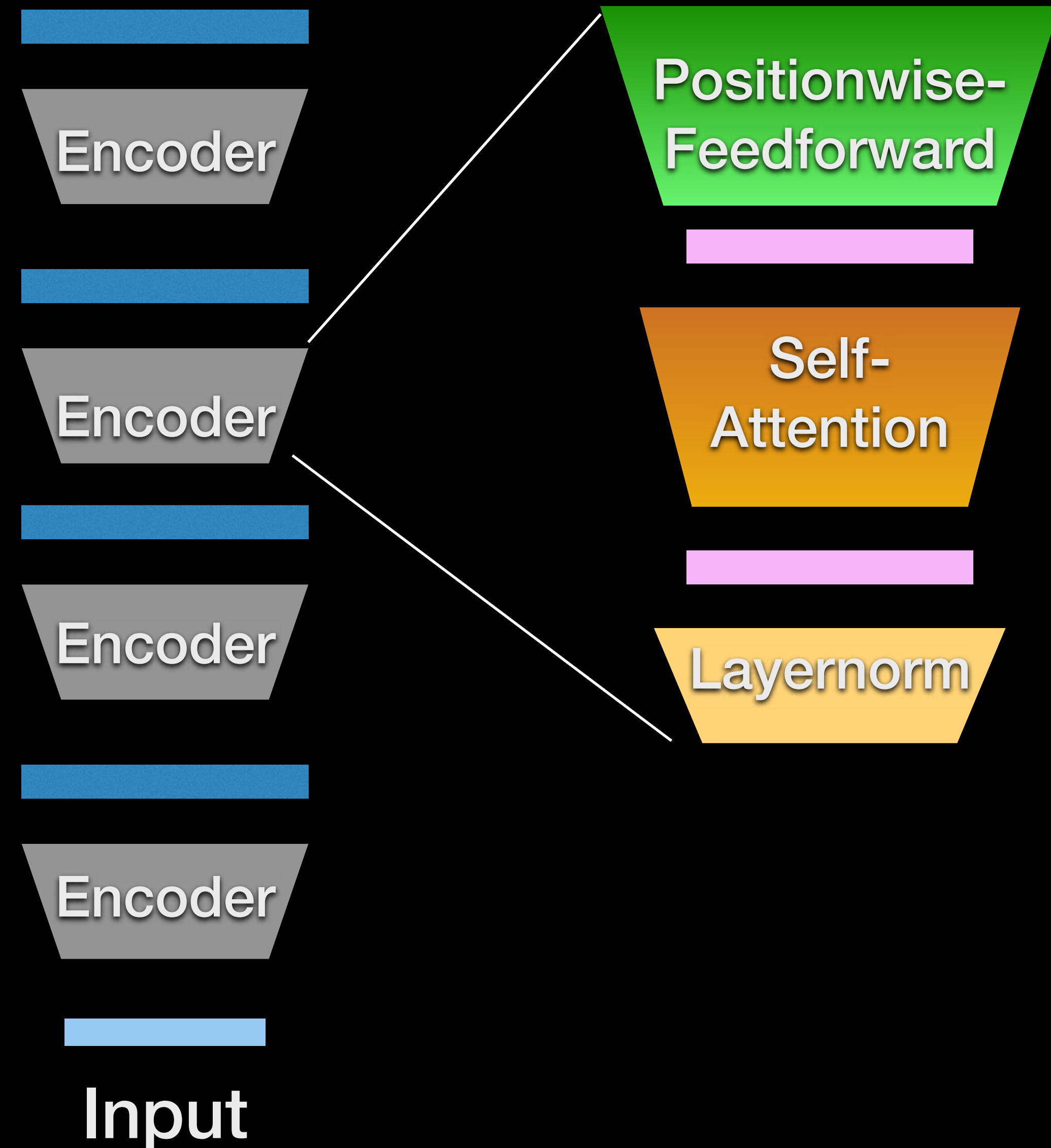
Transformers - the state of art in NMT



Transformers

✳ Encoder layers

- ➔ Consist of layer norm
- ➔ Self attention (multi-head)
- ➔ Positionwise feedforward
- ✓ May also consist of skip connections.



Transformers - encoder

✳ Let $\mathbf{x}(1) \dots \mathbf{x}(T)$ denote the input and let $\mathbf{e}^l(1) \dots \mathbf{e}^l(T)$ denote encoder outputs at layer l .

$$\overline{\mathbf{E}}^{l-1} = \text{Layernorm}([\mathbf{e}^{l-1}(1) \dots \mathbf{e}^{l-1}(T)]^T) \in \mathcal{R}^{T \times D}$$

✳ Definition of layer norm

$$\text{Layernorm}(\mathbf{e}^l(t)) = \frac{\alpha^l}{\sigma_{\mathbf{e}^l(t)}} \odot (\mathbf{e}^l(t) - \mu_{\mathbf{e}^l(t)}) + \beta^l$$



Transformers - encoder

* Query, Key and Value

$$\mathbf{Q}_h^l = \overline{\mathbf{E}}^{l-1} \mathbf{W}_h^{l,Q} + \mathbf{1}(\mathbf{b}_h^{l,Q})^T \in \mathcal{R}^{T \times d}$$

$$\mathbf{K}_h^l = \overline{\mathbf{E}}^{l-1} \mathbf{W}_h^{l,K} + \mathbf{1}(\mathbf{b}_h^{l,K})^T \in \mathcal{R}^{T \times d}$$

$$\mathbf{V}_h^l = \overline{\mathbf{E}}^{l-1} \mathbf{W}_h^{l,V} + \mathbf{1}(\mathbf{b}_h^{l,V})^T \in \mathcal{R}^{T \times d}$$

$$* \mathbf{W}_h^{l,Q}, \mathbf{W}_h^{l,K}, \mathbf{W}_h^{l,V} \in \mathcal{R}^{D \times d} \quad \mathbf{b}_h^{l,Q}, \mathbf{b}_h^{l,K}, \mathbf{b}_h^{l,V} \in \mathcal{R}^{d \times 1}$$

$$h = \{1..H\} \text{ heads} \quad d = \frac{D}{H} \quad \mathbf{1} \in \mathcal{R}^{T \times 1} \text{ all ones}$$



Transformers - encoder

* Multi-head attention

$$\hat{\mathbf{A}}_h^l = \mathbf{Q}_h^l (\mathbf{K}_h^l)^T \in \mathcal{R}^{T \times T}$$

$$\hat{\mathbf{A}}_h^l = \text{softmax}\left(\frac{\hat{\mathbf{A}}_h^l}{\sqrt{d}}\right)$$

$$\mathbf{C}_h^l = \mathbf{A}_h^l \mathbf{V}_h^l \in \mathcal{R}^{T \times D}$$

* Context vector from self-attention

$$\mathbf{C}^l = [\mathbf{C}_1^l \dots \mathbf{C}_H^l] \in \mathcal{R}^{T \times D}$$

Transformer - encoder

✳ Position wise feedforward layer

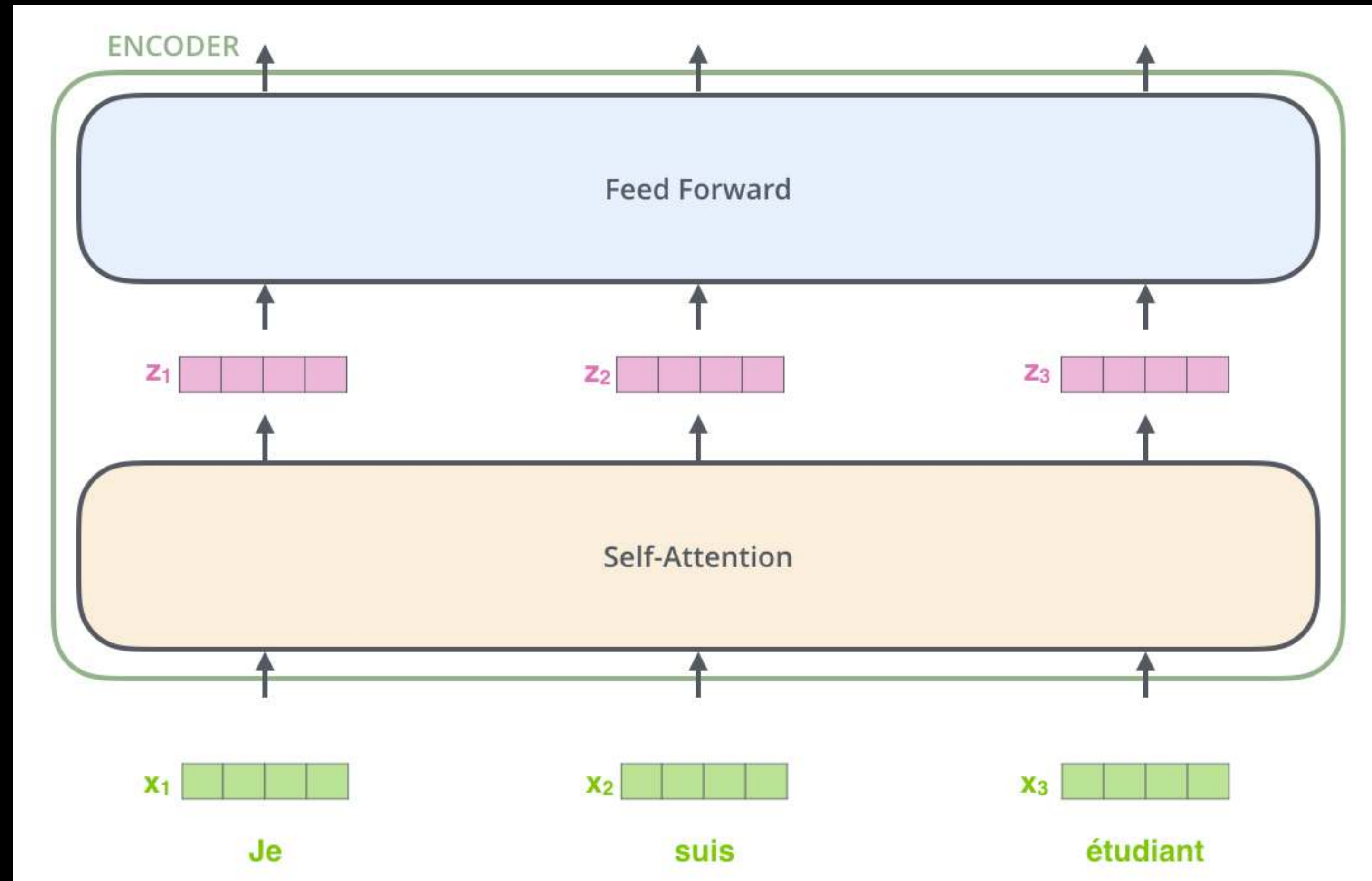
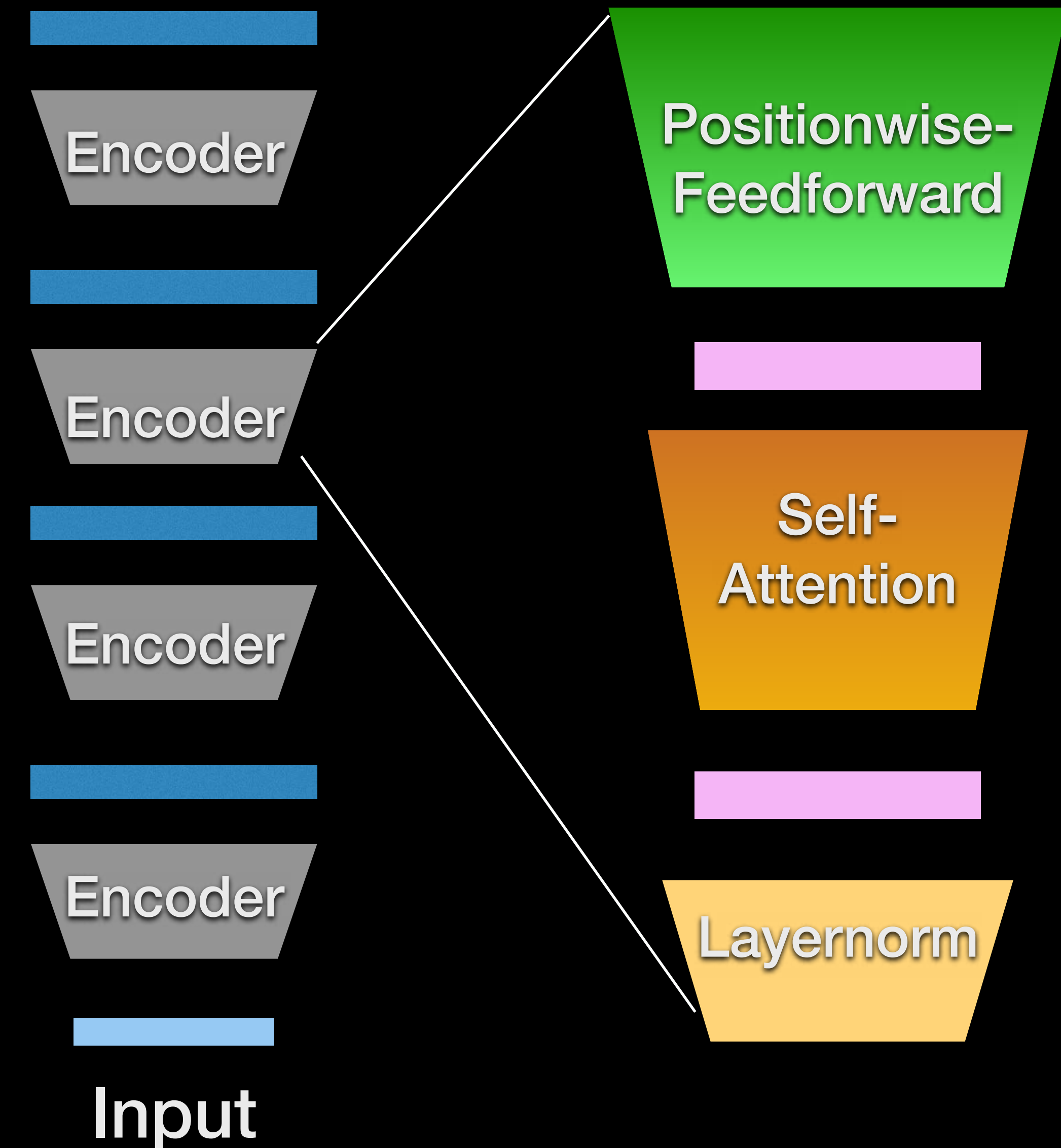
$$\mathbf{E}_{ff}^l = \text{ReLU}(\mathbf{C}^l \mathbf{W}_{ff}^l + \mathbf{1} \mathbf{b}_{ff}^T) \in \mathcal{R}^{T \times d_{ff}}$$

✳ Encoder layer output

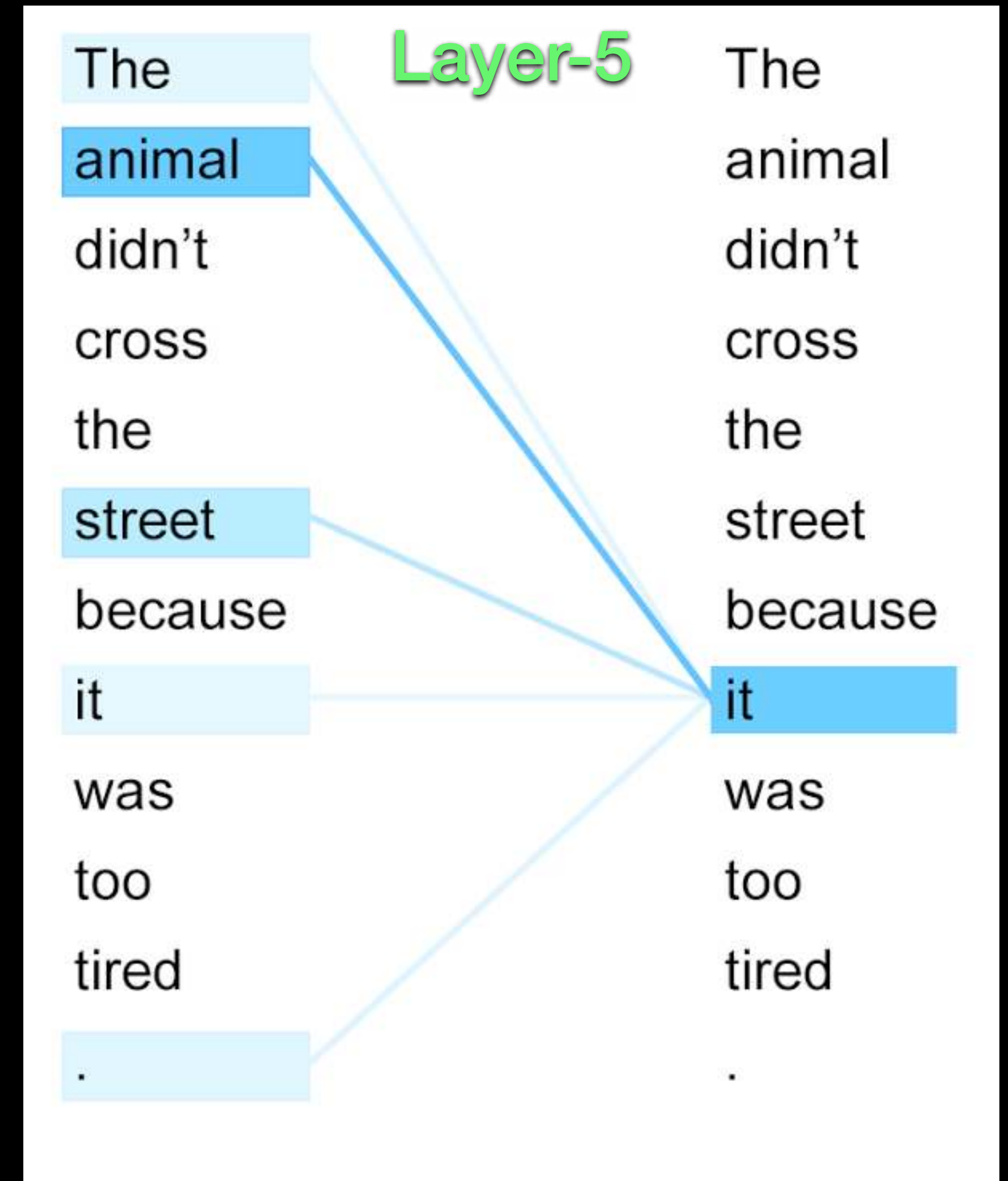
$$[\mathbf{e}^l(1) \dots \mathbf{e}^l(T)] = \mathbf{E}_{ff}^l \mathbf{W}_{of}^l + \mathbf{1} (\mathbf{b}_{of}^l)^T \in \mathcal{R}^{T \times D}$$



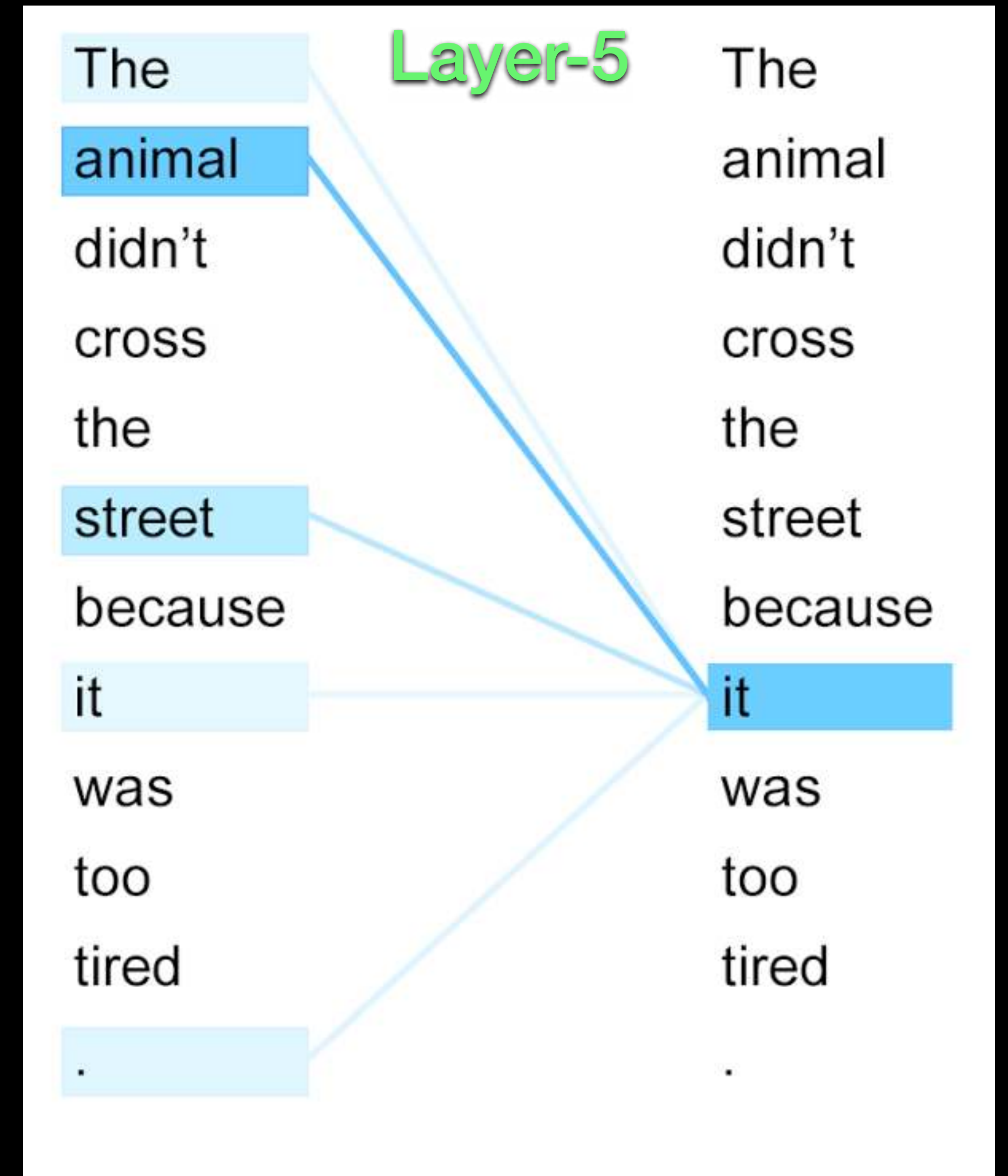
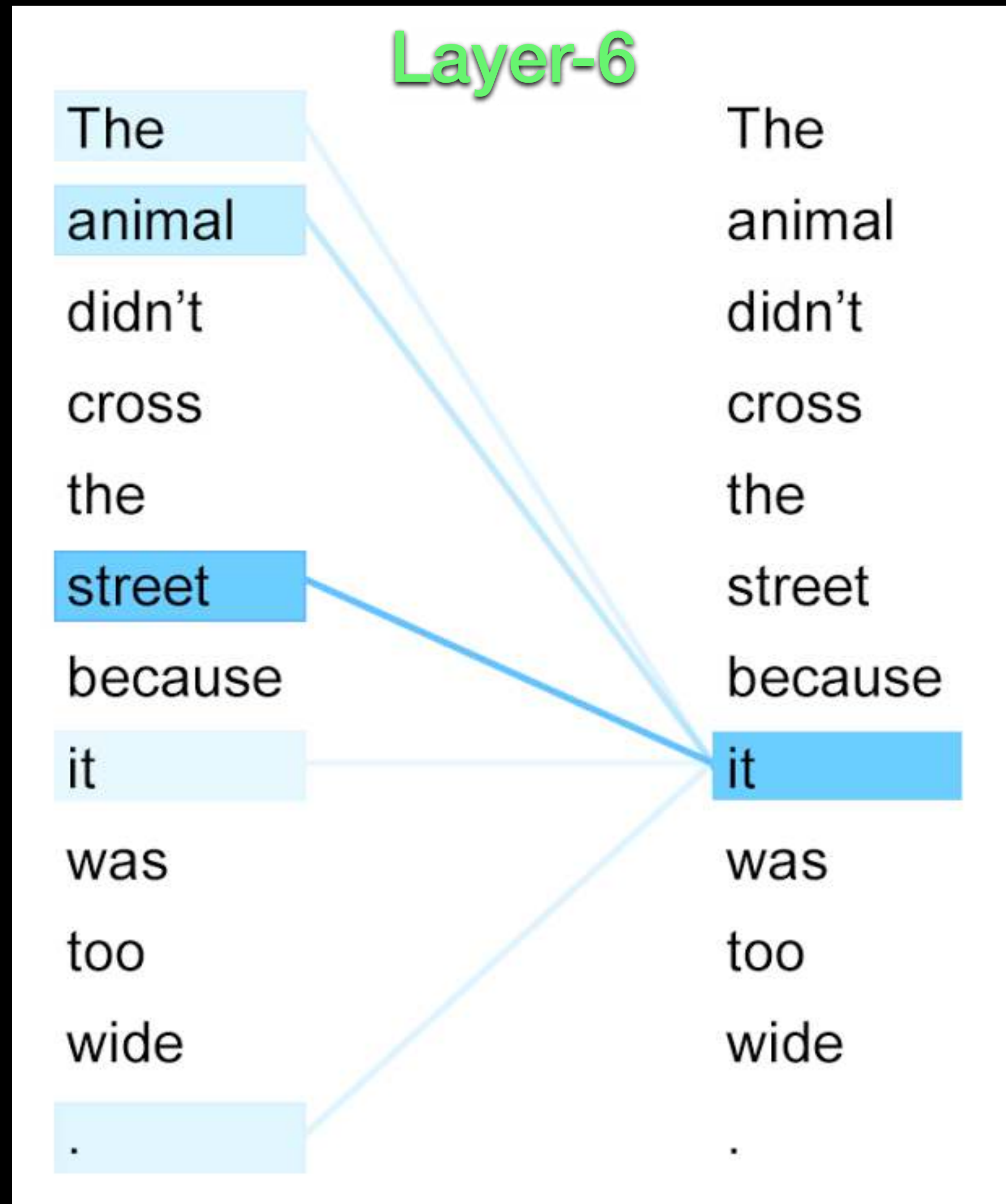
Transformers - encoder



Self-attention revisited

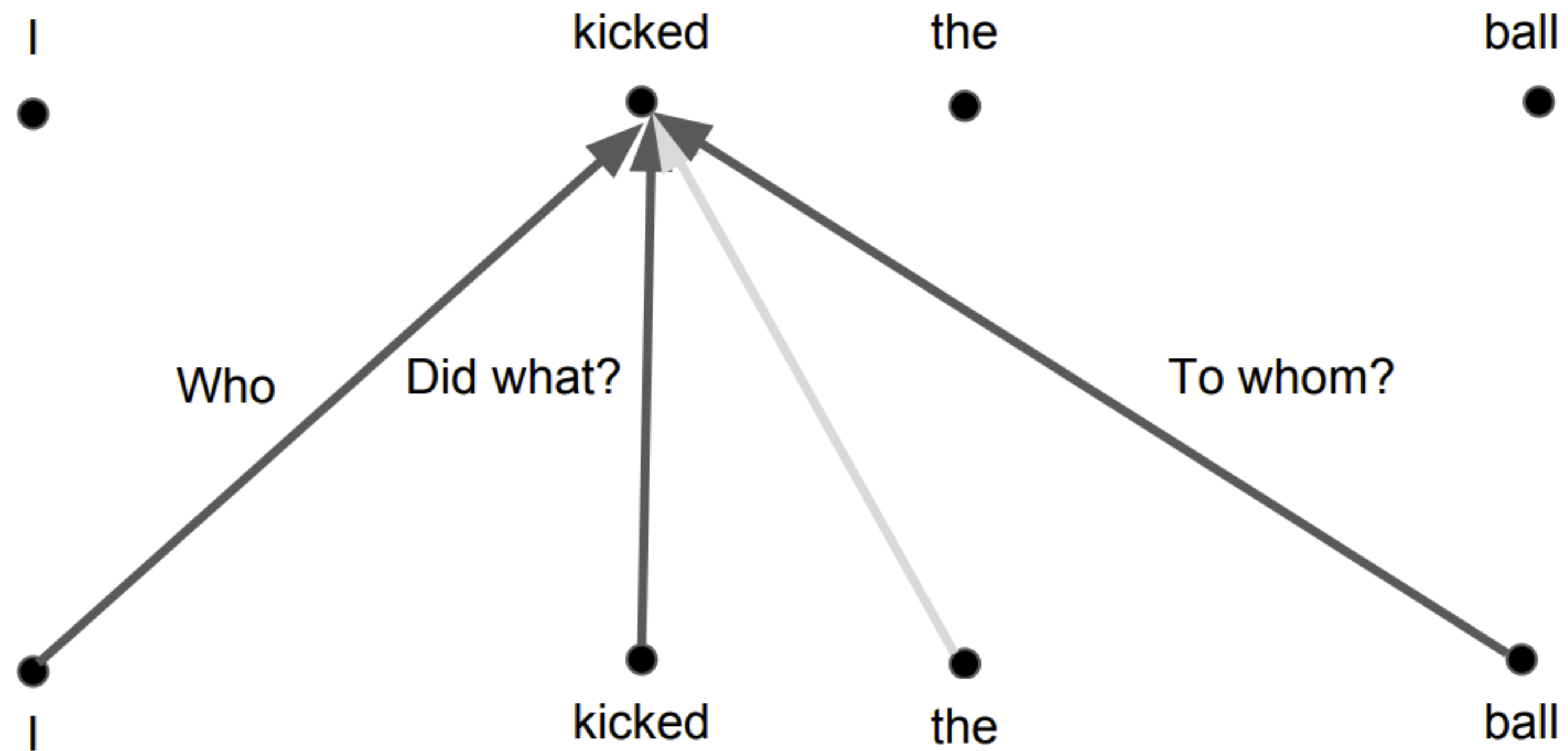


Self-attention revisited

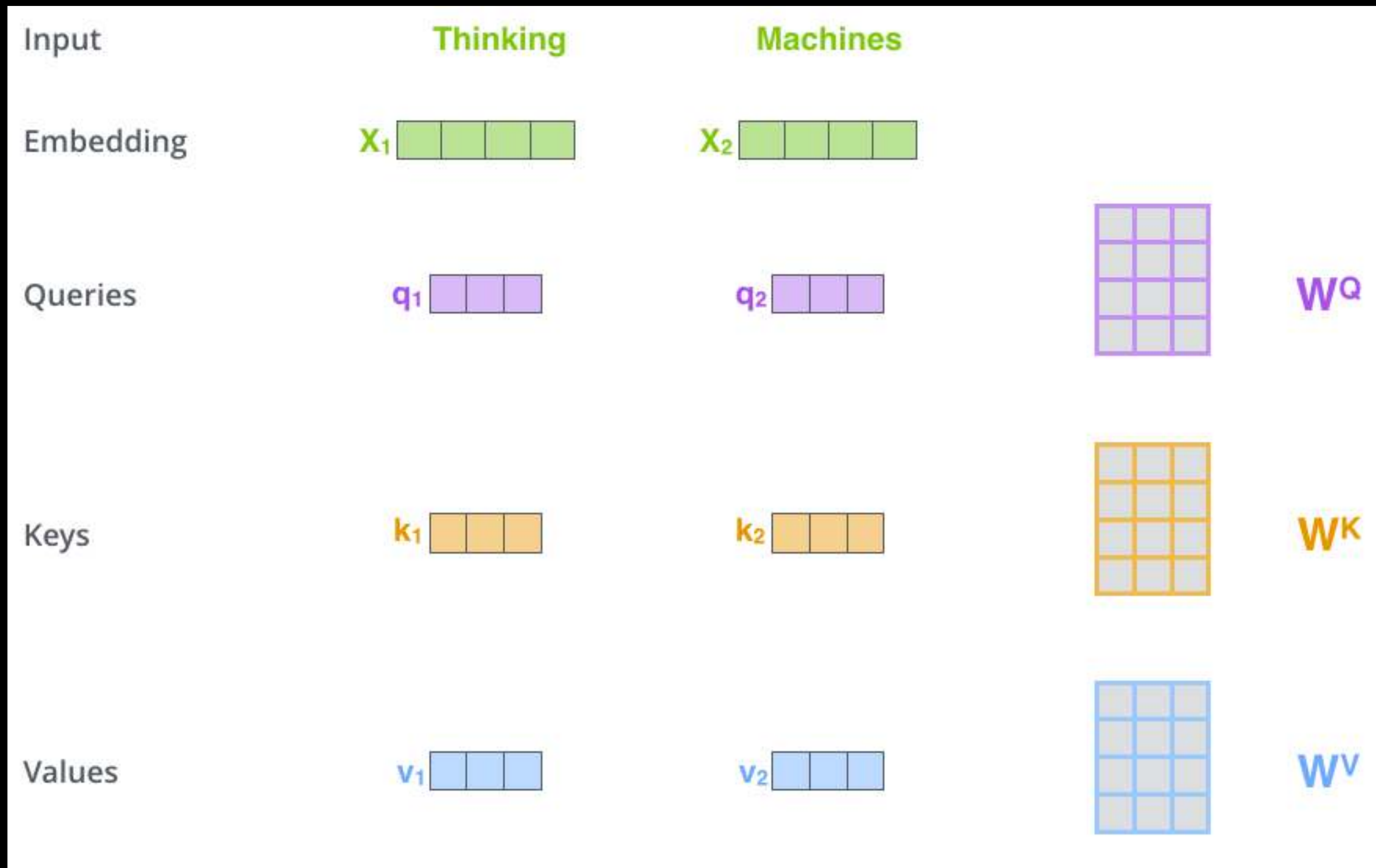


Self-attention revisited

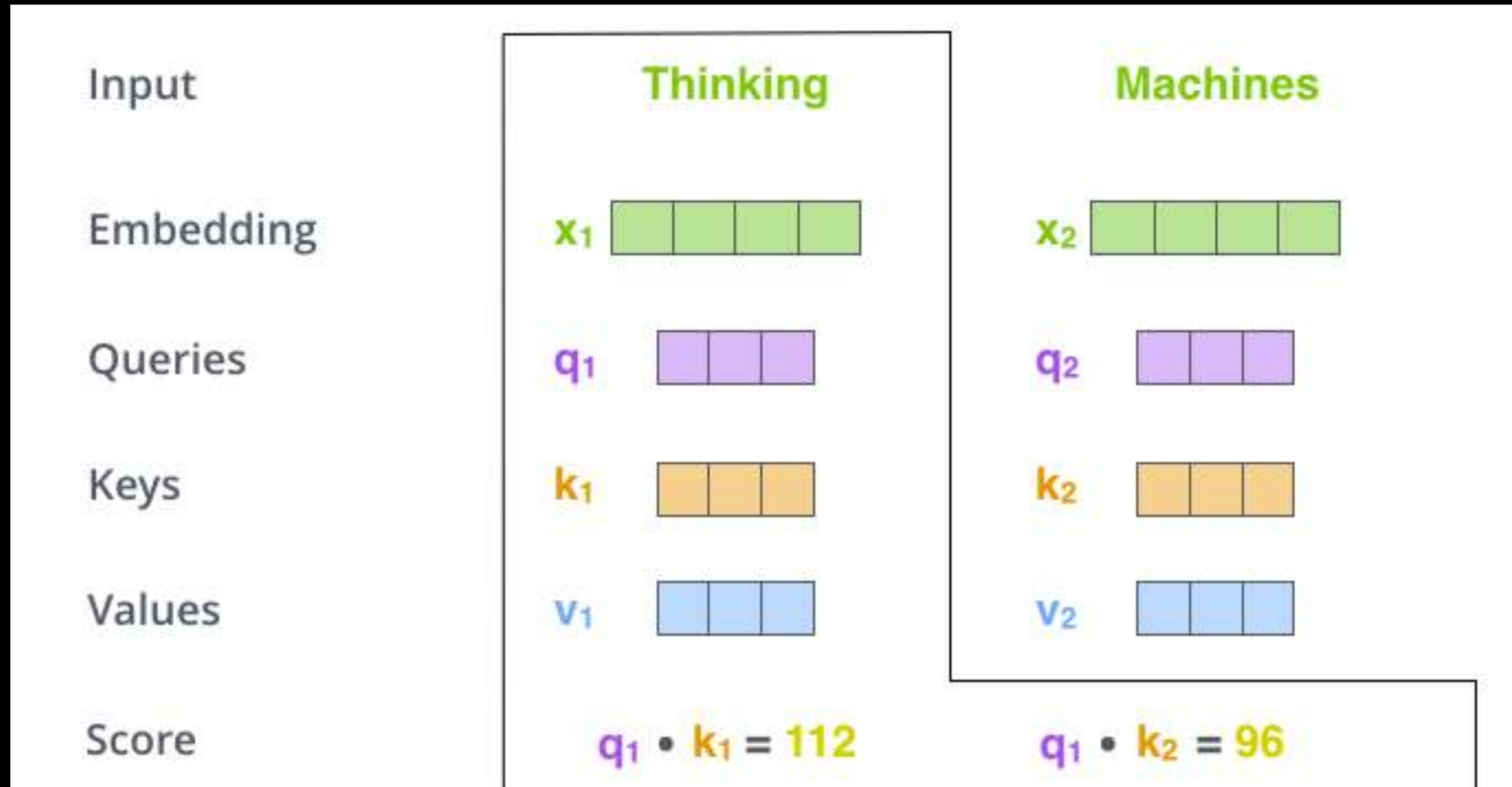
Self-Attention



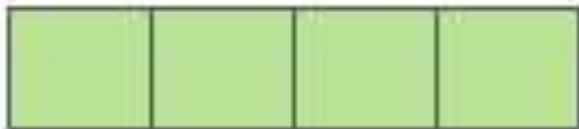
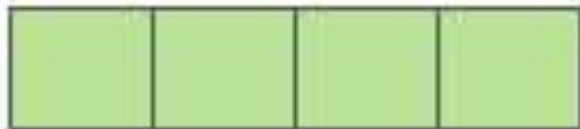
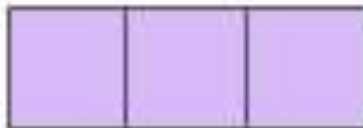
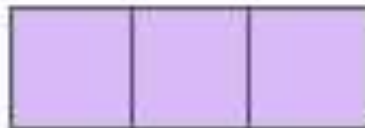
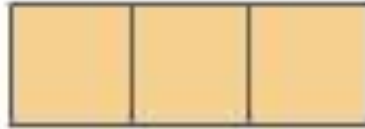
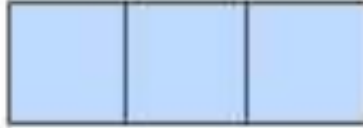
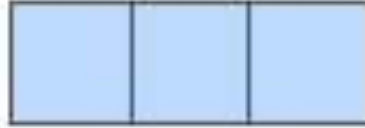
Self-attention revisited



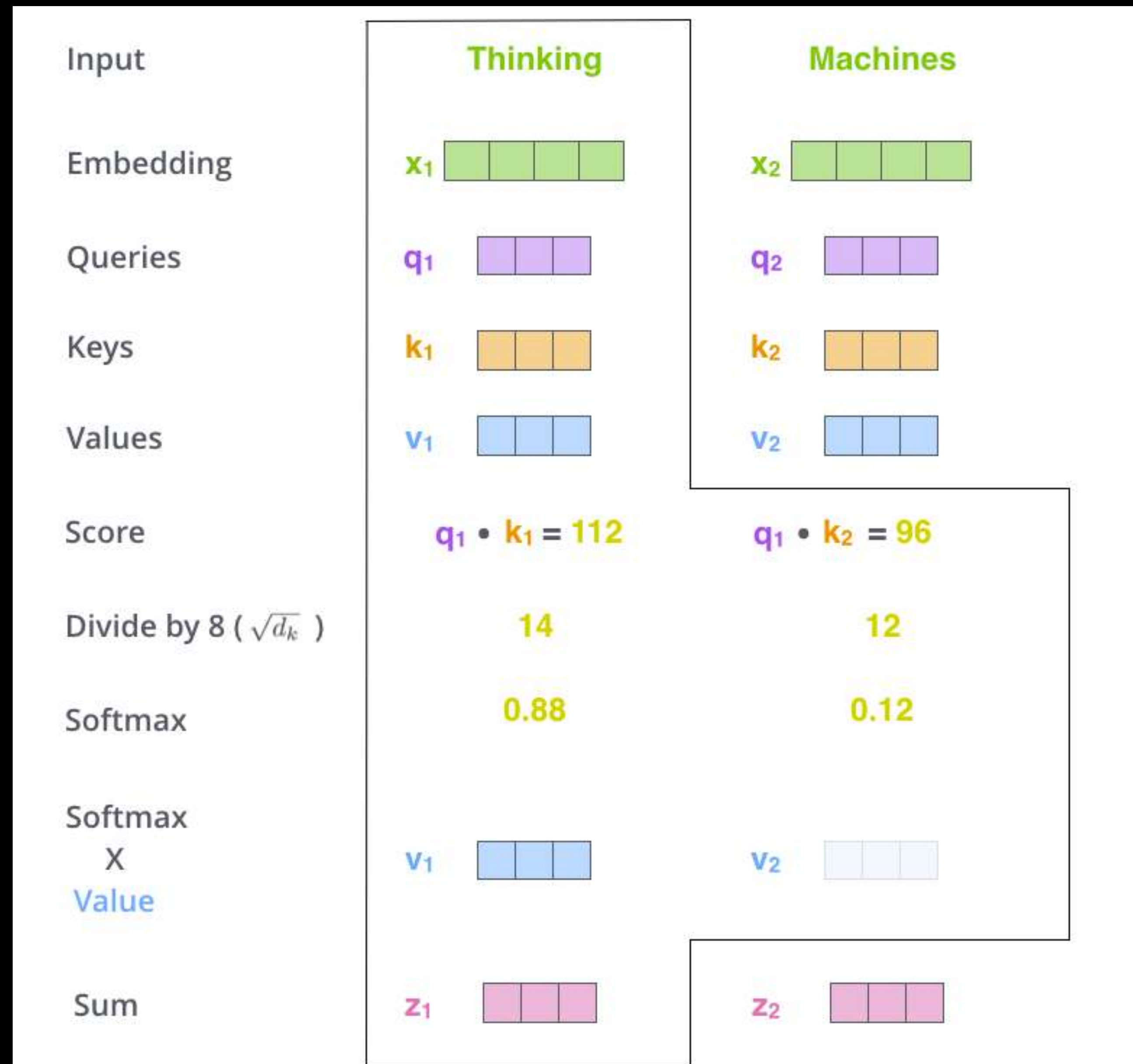
Self-attention revisited



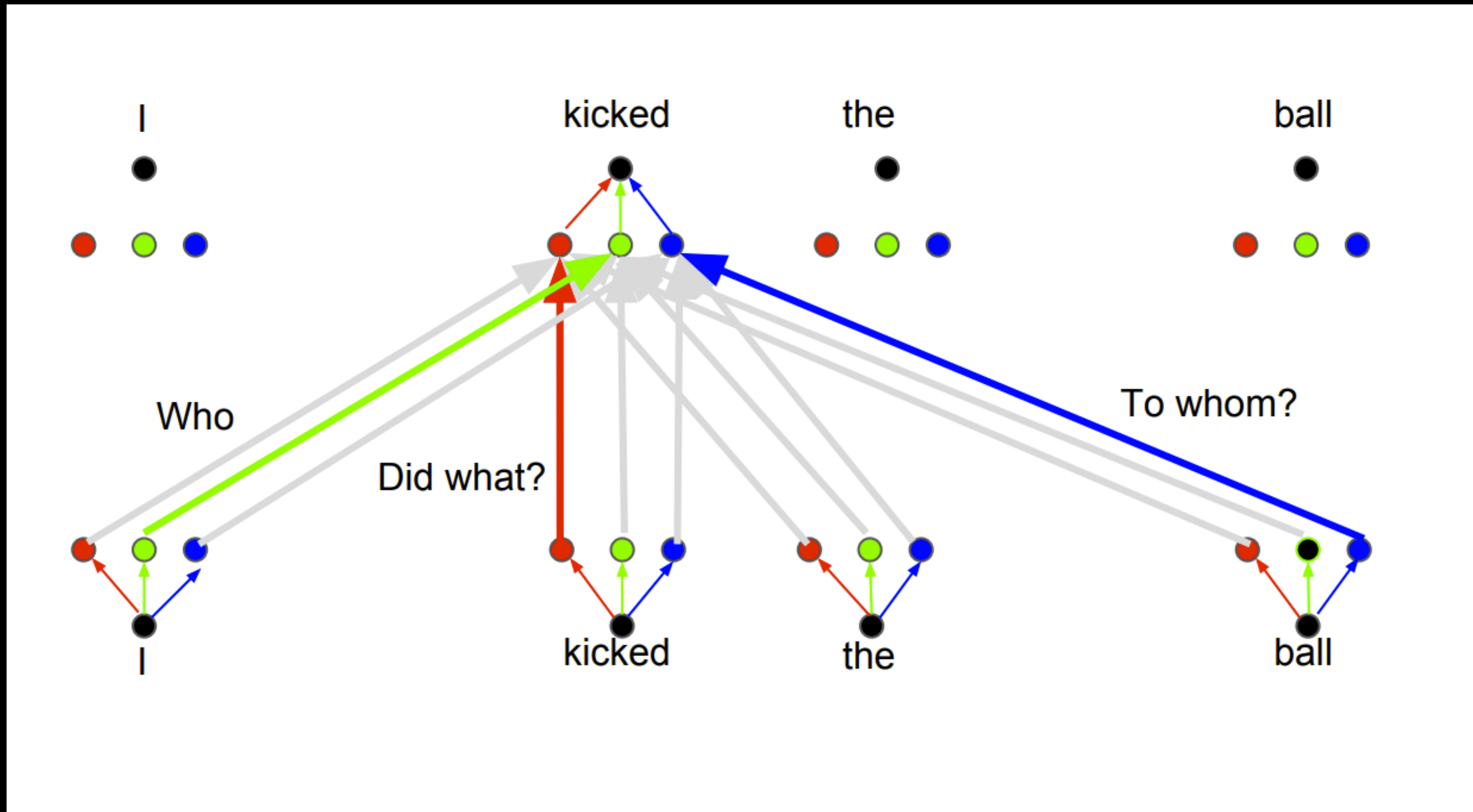
Self-attention revisited

Input	Thinking	Machines
Embedding	x_1 	x_2 
Queries	q_1 	q_2 
Keys	k_1 	k_2 
Values	v_1 	v_2 
Score	$q_1 \cdot k_1 = 112$	$q_1 \cdot k_2 = 96$
Divide by 8 ($\sqrt{d_k}$)	14	12
Softmax	0.88	0.12

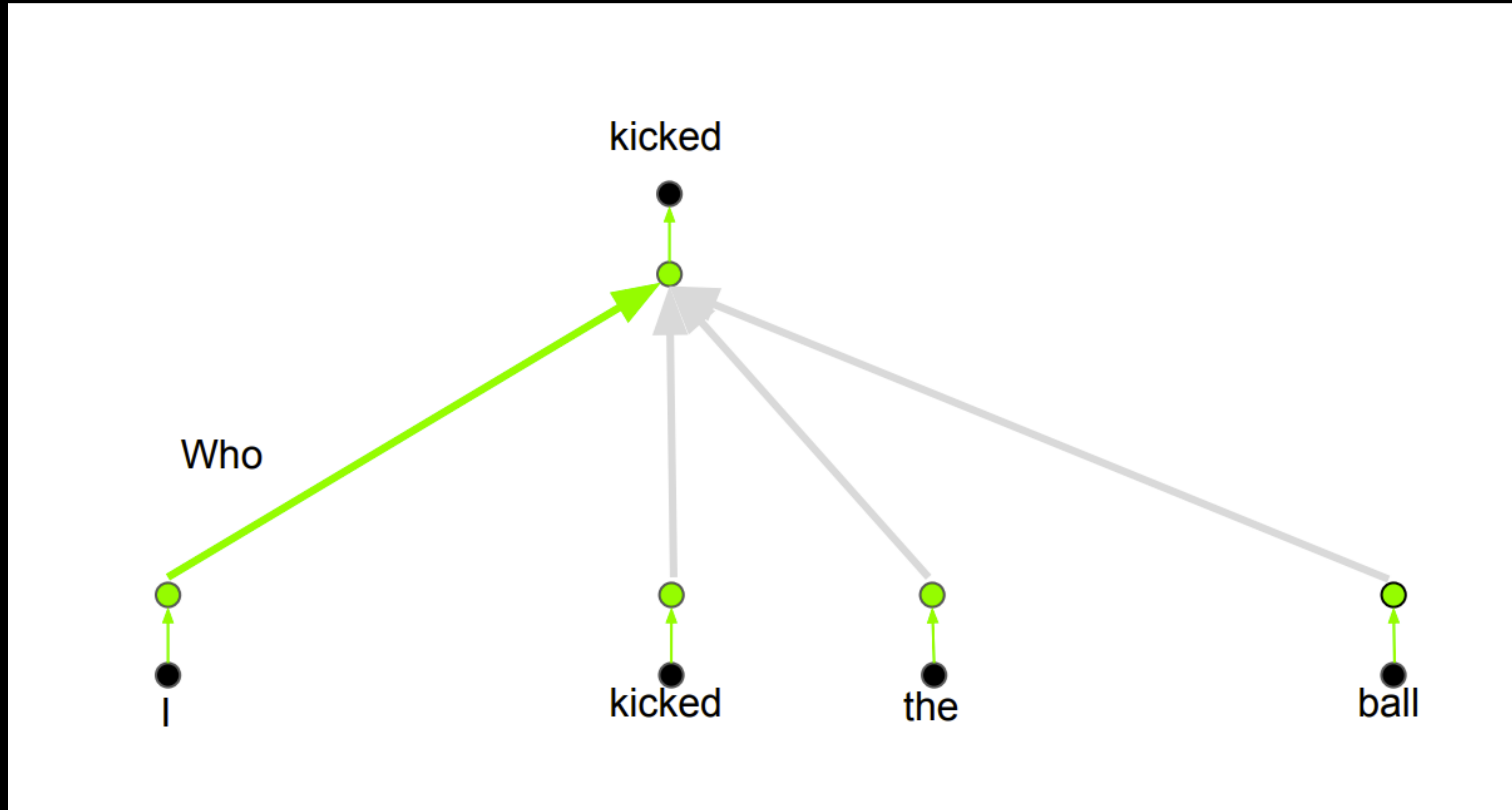
Self-attention revisited



Self-attention multi-head



Self-attention multi-head - role of attention heads



Self-attention multi-head - role of attention heads

